

School of Arts and Sciences
Department of Computer Science and Information Technology
Fall 2025 – 2026
CSCI373 Robotic Design and Coding

Class Day : _____ **Class Time** : _____ **Room**: _____
Instructor : _____
E-mail : _____
Office : _____ **Office Hours** : _____ **Extension No**: _____

I. Course Description

This course introduces the basic concepts and principles for using Arduino microcontroller platform as an instrument to teach students about topics in electronics, using C programming language, and human-computer interaction. Students will be able to build useful devices. Half of the in-class time is entirely devoted to developing, debugging, and refining projects where each session will have a period to solve a problem by the instructor and a period dedicated to the students to practice on a similar problem. This course can help you to ameliorate your programming skills, and to introduce you to hardware realization and application. At the end of the semester, pair or students or more will prepare a final project.

II. Course Objectives

- CO-1: Learn about components that plug into the main Arduino board.
- CO-2: Implement projects that utilize different capabilities of the Arduino.

III. Course Learning Outcomes

- CLO-1: Discover electronic and programming concepts.
- CLO-2: Develop a program to control the robotic components.
- CLO-3: Use Digital and Analog input/output.
- CLO-4: Test many robotic components (Resistor, LED, Push Button, Thermometer, photo resistor, Piezo, Seven-Segment LED Displays and Ultrasonic).
- CLO-5: Develop a program to use the robotic components then build a useful device or a robot to facilitate the real life.

IV. Mapping to program outcomes (PLOs are listed at the end of the syllabus)

A student who successfully fulfills the course requirements will have demonstrated the ability to:

Robotics Design & Coding	Program Outcomes								
Course Learning Outcomes (CLO)	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9
Discover electronic and programming concepts (10 %)	2X								
Develop a program to control the robotic components (25 %)				3X					
Use Digital and Analog input/output (15 %)		2X							
Test many robotic components (Resistor, LED, Push Button, Thermometer, photo resistor, Piezo, Seven-Segment LED Displays and Ultrasonic) (15 %)		3X							
Develop a program to use the robotic components then build a useful device or a robot to facilitate the real life (40 %)				3X			2X		

V. Topics Covered

- ☒ Exploring the Arduino Board and the IDE
- ☒ Building Blocks
- ☒ Working with Functions
- ☒ Numbers, Variables, and Arithmetic
- ☒ Motors and Movement

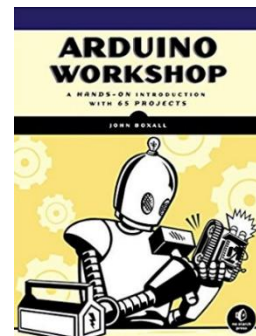
VI. Prerequisite – Corequisite

Course Prerequisite: CSCI250 - Introduction to Programming

Course Corequisite: None

VII. Course Material: Book or handouts

Arduino Workshop: A Hands-On Introduction with 65 Projects 1st Edition
by John Boxall – **Publisher:** No Starch Press; **ISBN-10:** 1593274483



VIII. Software required (for computer-lab courses)

Arduino.cc version 2.2.1, CH341SER (for Chinese board)

IX. Course Assessment

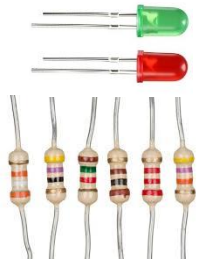
Midterm	35 %	Week 7
Final Exam	35 %	Week 14
Project	30 %	Week 15

X. Detailed Schedule

Book Reference: Arduino Workshop: A Hands-On Introduction with 65 Projects by John Boxall.

Week		Topic Details
Week 1 Sep 29 – Oct 02	Lecture 1	<u>Chapter 01.</u> Getting Started (Page 1) What is Arduino and why do we use it Reading Syllabus Parts and Accessories
	Lecture 2	Learn how to install the Arduino Software IDE or Tinkercad simulator and overview of the kit ❖ Required Software and Drivers ❖ Overview of the components in the Kit's components <u>Chapter 02.</u> Exploring the Arduino Board and the IDE (Page 20) Explain the Arduino board and the Arduino Software IDE or Tinkercad simulator ❖ The Arduino Board ❖ Taking a Look Around the IDE
Week 2 Oct 6–9	Lecture 1	❖ Blinking project: ▪ From IDE: File → Examples → 01. Basics → Blink ▪ https://www.arduino.cc/en/Tutorial/Blink ❖ Creating Your First Sketch in the IDE ▪ Comments ▪ The Setup Function ▪ The Loop Function ▪ Uploading and Running Your Sketch ▪ Modifying Your Sketch to blink two or multiple LEDs



	Lecture 2	<u>Chapter 03. First Steps (Page 33)</u> Explain about Electricity and Electronic Components ❖ Planning Your Projects ❖ About Electricity: Current - Voltage - Power ❖ Electronic Components ▪ The Resistor: Resistance, Reading Resistance Values, Chip Resistors, Multimeters, Power Rating ▪ The Light-Emitting Diode ❖ Fading project: ▪ From IDE: File → Examples → 01. Basics → Fade (https://www.arduino.cc/en/Tutorial/Fade)
Week 3 Oct 13–16	Lecture 1	❖ Project #01: Creating a Blinking LED Wave with Five LEDs (Page 43) ❖ Project #02: Repeating Project #01 with for Loops (Page 46) ❖ Project: Blink without delay (File → Examples → 02. Digital → BlinkWithoutDelay) (https://www.arduino.cc/en/Tutorial/BlinkWithoutDelay)
	Lecture 2	Project #03 on Pulse-Width Modulation and Explain more Electronic Components ❖ Varying LED Brightness with Pulse-Width Modulation ❖ Project #03: Demonstrating PWM (Page 49) ❖ More Electric Components ▪ The Transistor - The Rectifier Diode - The Relay ❖ Higher-Voltage Circuits 
Week 4 Oct 20–23	Lecture 1	<u>Chapter 04. Building Blocks (Page 55)</u> Explain Schematic Diagrams ❖ Using Schematic Diagrams ▪ Identifying Components: Arduino - Resistor - LED - Transistor ▪ Wires in Schematics ❖ The Capacitor ❖ Explain Digital Inputs and do Project #04 ❖ Digital Inputs ❖ Project #04: Demonstrating a Digital Input (Page 65) ▪ Creating Constants with #define ▪ Reading Digital Input Pins ▪ Making Decisions with if 
	Lecture 2	❖ Boolean Variables (Page 72) ❖ Extra projects with push buttons (digitalRead, digitalWrite, analogWrite...)
Week 5 Oct 27–30	Lecture 1	The photo resistor (22 LDR SMALL) ❖ Display the value of the photo resistor in the serial monitor. Project: LED Turn on in the Dark. 
	Lecture 2	❖ Extra projects for photo resistor

Week 11 Dec 8–11	Lecture 1	Ultrasonic Distance Sensors ❖ Ultrasonic Distance Sensors ❖ Display the value of the ultrasonic in the serial monitor 
	Lecture 2	❖ Project: Make Car parking system using the ultrasonic with two LEDs and one piezo. ❖ Extra projects
Week 12 Dec 15–18	Lecture 1	Chapter 12. Motors and Movement (Page 225) Explain Servo Motor ❖ Making Small Motions with Servos ❖ Sweep (https://www.arduino.cc/en/Tutorial/Sweep) ❖ Project #38: Building an Analog Thermometer (Page 228) or any extra project 
	Lecture 2	❖ Extra Exercises
Week 13 Dec 22–25	Lecture 1	Projects presentation on Mon 22 and Tue 23 Dec 2025
	Lecture 2	Holidays (Wed 24 and Thu 25 Dec 2025)
Week 14 Dec 29 – Jan 01	Lecture 1	Holiday (Mon 29 and Tue 30 Dec 2025)
	Lecture 2	Holiday (Wed 31 Dec 2025 and Thu 1 Jan 2026)
Week 15 Jan 5–8	Lecture 1	Holiday (Tue 6 Jan 2025) Projects presentation on Mon 5 Jan 2026: ❖ For TTh sections: during office hours ❖ For MW sections: in the Lab
	Lecture 2	Final Exam (Wednesday 07 Jan or Thursday 08 Jan). Topics Included: everything in Midterm focus on piezo, Seven-Segment Display, Ultrasonic and Servo Motor

Class Rules & Discipline

Attendance Policy

The student is administratively withdrawn (AW) from the class if he/she is absent for more than one third of the classes (15 class sessions for courses offered 3 times a week, 10 for courses offered twice a week, and 5 for courses offered once a week). If the number of absences exceeds the limit after the withdrawal deadline, i.e. including all the period up-to the end of the semester, the student will not receive an AW grade, therefore, the final examination must be written, otherwise, an F grade will be granted. When absent, you are responsible for all material presented in class. Missed exams will only be excused by documented evidence as explained in the make-up examination policy below.

A “10%” has been allocated on the “Marking Profile” for attendance. A student who hasn’t missed any lecture will get a full mark on the attendance grade. Every absence, a 20% will be deducted, that is, a student with 5 absences will get a ZERO on that grade.

Cheating

The University policy on cheating will be strictly applied. Students are required to adhere to the policy on academic irregularities contained in the university student handbook. Collaboration among students in solving assignments is not acceptable. It is okay to ask another student for advice when stuck on a problem, but if the instructor receives assignments that are substantially equivalent, all involved students will be punished with a grade of zero.

Classroom Manners

All students are expected to conduct themselves in a professional manner. Students should not: keep their cell phones on during the class, disrupt the class, do other course work in class, have side conversations with other students, be rude towards anyone. The first failure to respect these rules will result in a *warning* from the instructor. If a student chooses to disregard the *warning*, he/she will be dismissed from class.

Instructor late to the class

In case the instructor is late the students should wait for 15 minutes before leaving the class.

Abuse of Computing Privileges

The Office of Information Technology maintains policies posted in the computer labs regarding proper behavior in the labs. Failure to adhere to these policies may result in loss of computer privileges, and possible legal action.

Make-Up Examination Policy

As per university council decision, a student is eligible for a mid-term or final examination make-up if and only if he/she had the following incidents:

- a. Sickness; **proved by hospitalization report**; that is; a discharge summary is necessary.
- b. Death in the family **proved by a death certificate** and personal identification.
- c. Accidents **proved by an expert report**.

A student that misses an exam should submit a petition within one week from the examination date. Late petitions will be rejected unless it is an extreme case (Hospitalization as an example).

In case a petition is accepted, the student submits a MAKE-UP exam. The grade of the missing exam will be equal to a percentage % of the Final Exam’s grade. The percentage % will be set by the administration once the petition is approved.

Cheating policy: Any cheating will be rewarded with a zero, all students involved in cheating will get a zero (even if the cheating is in one question). The activity would not be repeated.

Missing Assignments and Projects Policy

- A -10 per day penalty will be applied to late projects (weekends count as two days).
- A ZERO will be applied for late assignments on the next day.

Curriculum, Yearly Plan, and Advising

If a student majoring in Computer Science or Information Technology is planning to graduate in three years then he/she should follow the corresponding major yearly plan. A copy of the yearly plan and the curriculum (which contains the prerequisites and co-requisites of courses) can be picked up from the office of the school of Arts and Science. When not sure about anything please contact your advisor prior to or during the registration period.

Probation Policy

If a student who has attempted 19 credits has a GPA or a major GPA below 2.0 then the student is considered under probation.

E-mails

Students should use their university e-mails in communicating with the instructors. Important announcements, syllabi, assignments, and solutions will be sent to your university e-mail accounts.

University Peer Review Policy

All classes will be visited by instructors either from the same or different departments. This policy helps instructors figure out ways to improve student educational experiences in class and eventually their performance.

Snapshots

Students are not allowed to take snapshots of the whiteboard in classrooms using their mobile-phones.

Equivalent Lettering Grades

<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>	<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>
<60	0	F	67	1.7	C
60	1	D	68	1.8	
61	1.1		69	1.9	
62	1.2		70	2	
63	1.3		71	2.1	
64	1.4		72	2.2	
65	1.5	D+	73	2.3	
66	1.6		74	2.4	
75	2.5	C+	83	3.3	B+
76	2.6		84	3.4	
77	2.7		85	3.5	
78	2.8		86	3.6	
79	2.9		87	3.7	
80	3	B	88	3.8	
81	3.1		89	3.9	
82	3.2		>=90	4	A

<u>Grade</u>	<u>Description</u>	<u>QPTS</u>
A	Excellent	4.00
B+	Very Good	3.50
B	Good	3.00
C+	Above Average	2.50
C	Average	2.00
D+	Fair	1.50
D	Poor	1.00
F	Fail	0.00
IP	Course in Progress	
I	Incomplete work	
W	Student has withdrawn from the course	
AW	Administrative withdrawal from the course	
R	Course has been repeated	
S	Satisfactory work	
U	Unsatisfactory work	
%G	Final Grade	
QPTS	Quality Points	

Computer Science Outcomes

Graduates of the Computer Science Program will have the ability to:

Program Outcomes	Computer Science Outcomes Graduates of the Computer Science Program will have the ability to:
Category -1 Underlying Conceptual Basis for Informatics	PO-1 Describe and explain the essential facts, concepts, theories and mathematical methods relevant to computing, computing equipment, computer communication and informatics applications as appropriate to their program of study
	<i>PO-1 for CSIT=> Gain basic skills in mathematics and outline the characteristics of relevant state-of-the-art hardware and software and their practical application</i>
Category -2 Analysis	PO-2 Select and use relevant analytic, modelling and simulation methods
	PO-3 Choose appropriate solution patterns, algorithms and data structures
Category -3 Design and Implementation	PO-4 Describe the software development process and apply relevant practical and programming skills to the creation of computer programs, and/or other informatics artefacts
	<i>PO-4 for CSIT=>specify and design computing/network hardware/software which meet specified requirements</i>
Category -4 Economic, legal, social, ethical and environmental context	PO-5 Explain the importance of information privacy and security issues in relation to the design, development, maintenance, monitoring and use of informatics-based systems
Category -5 Informatics practice	PO-6 Describe and explain management techniques appropriate to the design, implementation, testing, deployment and maintenance of informatics systems, including project management, configuration management, change management, etc., and including relevant automated techniques
	<i>PO-6 for CSIT=> demonstrate an awareness of project management and business practices, such as risk and change management, and understand their limitations;</i>
Category -6 Other Professional Skills and Competences	PO-7 Communicate effectively both verbally and using a variety of communications media
	PO-8 Plan self-learning and improve personal performance as a foundation for lifelong learning and ongoing professional development
	PO-9 Participate effectively in informatics group-working